# Energy communities: close the gap with local operations

Public energy-poverty data, the real cost of rural OPEX, and cooperatives as a sustainability vehicle for utilities and foundations in Colombia.



Photo: Alexander Mass / Pexels (free licence)

## 1 The national gap

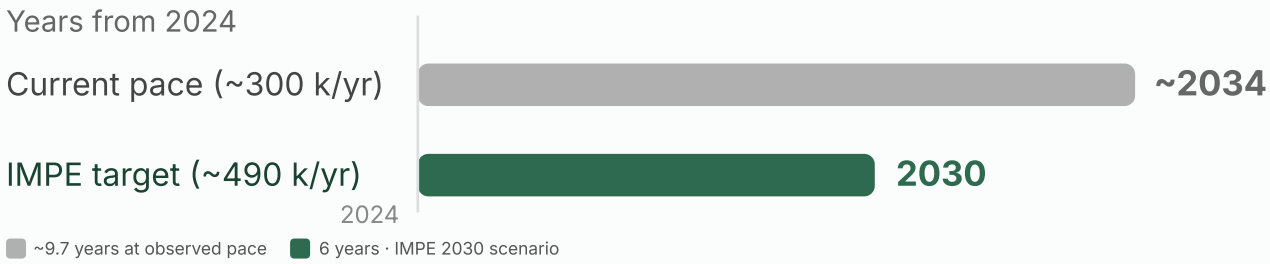
**15.4%**
IMPE incidence 2024 (~8.1 M people)[1]

**9.5%**
2030 scenario target (~5.17 M; −2.9 M)[1]

**75.9% / 93.1%**
Rural vs national residential coverage (ICEE 2024)[2]

**63.1% / 52.8%**
IMPE La Guajira / Chocó (2024)[1]

**When does the 2.9 M-person gap close?[1]**




## 2 OPEX, not just CAPEX

**Contraloría**
No methodology to pay for O&M of microgrids and PV in remote areas.[3]

**Amazonas**
~$30,000 M (royalties) in PV not operating; damaged batteries — abandonment risk.[4]

**CREG**
Thousands of rural PV systems with no management → often abandoned.[5]

**CTH differentiator:** we do not compete on MW funds or greenfield; we make the social and OPEX layer *bankable* where an executor and CAPEX are already committed.

[1] IMPE Report 2025 (2024 base), pp. 12, 22, 33. [2] UPME ICEE Bulletin 2024. [3] Contraloría, remote solar projects. [4] Seguimiento.co / Procuraduría, Amazonas. [5] CREG GEN-Renewables report; Concept 4022/2019 (O&M).

## 3 Cooperatives as the tool

CTH PROPOSAL **Community O&M + revenue + local ownership.** OPEX contracts (PV, batteries, monitoring; CREG/SSPD framework) and cooperative governance. KPIs: IMPE access/quality dimensions — not a substitute for ICEE coverage.

## 4 Ecosystem startups

**SN**
**Sunco (SunColombia)**
Bogotá
Design, build, and O&M for on/off-grid solar for institutions and non-interconnected zones.
**898 kWp · 1,122 users (DISPAC, La Guajira)**[11]
suncolombia.co

**SE**
**SEOS Energy**
Bogotá
Platform linking homes and SMEs to certified installers and solar financing.
**75+ SOLvers · 18 departments**[12]
seosenergy.co

**SL**
**Soluna Energía**
La Guajira · Vichada
Prepaid pay-as-you-go solar for off-grid households in the Caribbean and Orinoquía.
solunaenergia.com

**TG**
**Tierra Grata**
Cartagena
Decentralized energy and microfinance in remote rural communities (PayGo model).
**1,947 clean energy solutions**[13]
tierragrata.org

**TW**
**TuWatt**
Bogotá
Structures and operates distributed generation and energy-community projects with traceability.
tuwatt.co

**Local talent:** Escuela Colombiana de Ingeniería Julio Garavito — 101.76 kWp lab, renewable diplomas, and energy-sector challenges (Solucionatón 10X).[6]

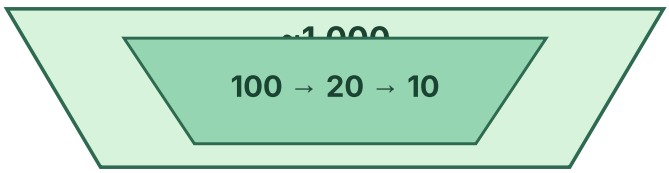
## 5 PASO Colombia

**Productive communities** in high energy-poverty departments. **PASO does not run energy projects.**[9]

- Fonseca, La Guajira (IMPE 63.1%)
- Puerto Asís / Guzmán, Putumayo (58.7%)
- Istmina, Chocó (52.8%)
- San José del Guaviare (38.9%)
- Arauca (27.9%)

## 6 Data lab

MinEnergía-prioritized communities → shortlist (solar resource, off-grid areas, legal status, social license).[9]




*Funnel figures: illustrative.*

**Energy Coop candidate community:** Corozal, Sucre (IMPE 43.1%) — not a PASO project.[9]

## 7 Partners and next step

FMO | RVO | Green Choice | Energie Samen

Solar finance and facilitation (FMO, RVO).[8] Peer learning on energy communities — Green Choice, Energie Samen (target; no signed agreements).[9]

**Next step:** co-design session with executor and candidate community — governance, OPEX, and M&E aligned to IMPE. cleantechhub.net

[6] escuelaing.edu.co — renewables lab and continuing education. [8] FMO and RVO — solar finance/facilitation (target partners). [9] CTH / PASO 2026, project brief. [11] suncolombia.co — DISPAC project; cleantechhub.net (SUNCO portfolio profile). [12] seosenergy.co. [13] tierragrata.org — clean energy impact.


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